



Hackathon: Building an Intelligent Domain-Specific AI Assistant with RAG Technology

2025



Hackathon Objective



CHALLENGE

**Design and develop a domain-specific intelligent
AI assistant leveraging Retrieval-Augmented
Generation (RAG) techniques**

Application Options



Healthcare Assistant

Medical guidance, symptom checking,
medication info



Financial Advisor

Investment advice, market insights, portfolio
optimization



Legal Consultant

Legal definitions, procedures, case law lookup



Education Tutor

Subject explanations, problem solving,
learning aids



Travel Planner

Custom itineraries, local recommendations,
booking help



Custom Domain

Any unique domain proposed by your team

AI Tools Usage Guidelines

Category	Technology Options
LLM APIs	Azure OpenAI API (GPT-4/4o), HuggingFace LLMs
Orchestration	LangChain, LangGraph (for multi-step/agent workflows)
Embeddings	HuggingFace Transformers for custom models/embeddings
Vector Database	Pinecone, Chroma
Frontend	Streamlit, Gradio (Python), React.js (JS), Flask/FastAPI



Pro Tip

Choose technologies your team is comfortable with - speed of development matters in hackathons!

Deliverables by End of Hackathon

 **User Stories & Use Case Documentation**

 **MVP Feature List with Implemented Functionalities**

 **System Architecture Diagrams**
Interface Mockups & Screenshots

 **Source Code Repository (GitHub)**
Deployed Application URL

 **Test Plan and Results Summary**
 **Team presentation with demo**



Evaluation Criteria

💡 **Innovation
& Use Case**

20pts

Originality and
domain
relevance

🔑 **RAG
Pipeline
Implementation**

25pts

Correctness and
efficiency of
retrieval &
generation

🎨 **User
Interface
Quality**

15pts

Usability,
design, and
interactivity

🎯 **LLM
Response
Relevance**

15pts

Accuracy and
contextual
relevance

✏️ **Testing &
Robustness**

10pts

Handling edge
cases and
errors

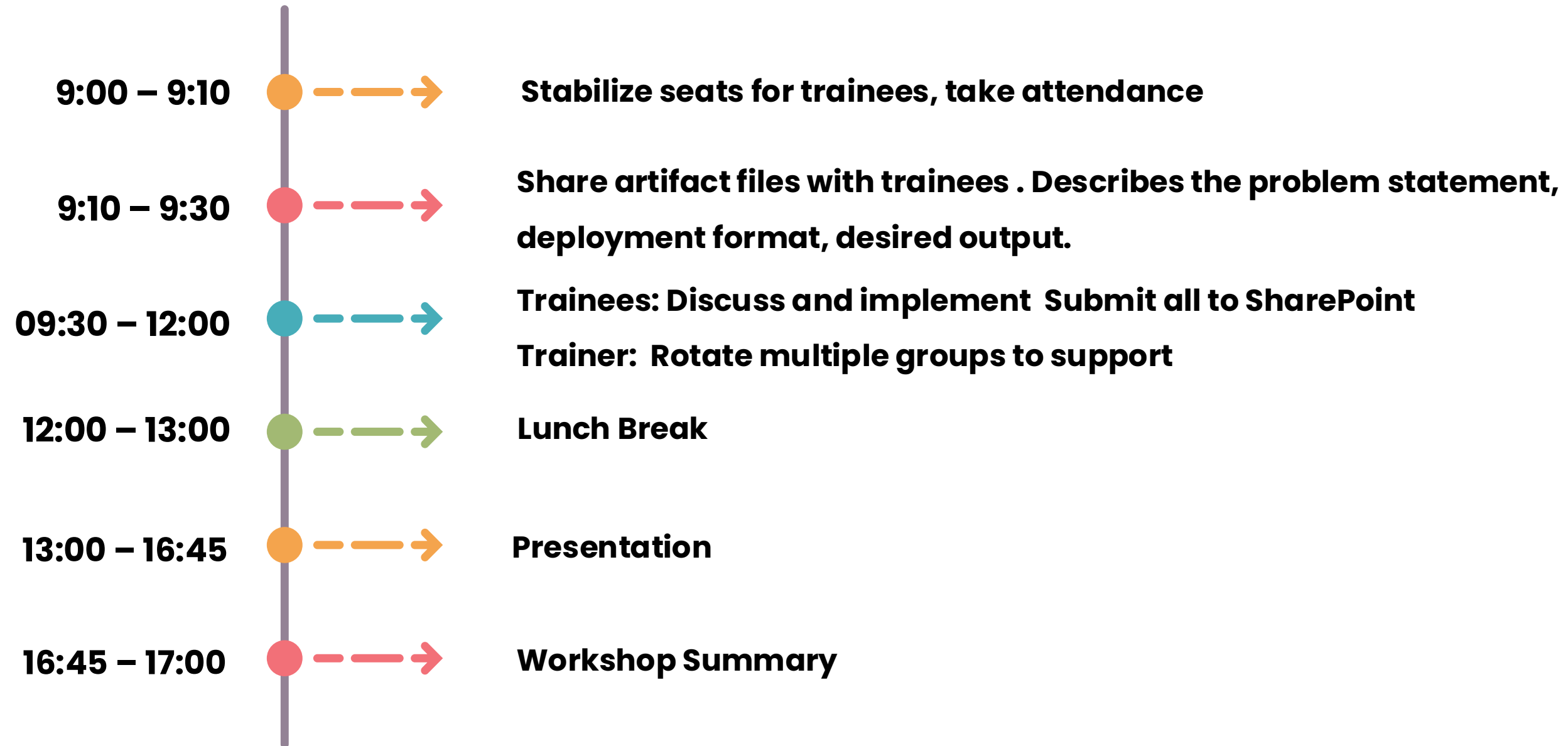
🚀 **Deployment
& Demo**

15pts

Working
deployment and
presentation
quality



Timelines



THANK YOU!

 **Ready to Build Amazing AI Tools?**
Let's turn your repetitive tasks into automated solutions!